

The top-mode edge

–sch = $\beta - \alpha$ is the r401 chart rewrite; the barycentric hole-top mode is not the bulk defect

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Abstract

After rounds 407–411 the reviewer (DCCLXXIX) asks that the edge half of P1 be re-read as a *top-mode balance*: if the hole-top-mode theorem holds in the bulk, the only critical direction is the barycentric mode $v_{\text{top},i} \propto 1/(\sqrt{u^\vee(y_i)} P'_Y(y_i))$, and $-\text{sch}_k = \beta_k - \alpha_k$ with α the defect energy after bulk continuation and β the net CD lift after border consumption. Round 405 tested the constant (the wrong vector); the Euler tail and the disk Parseval identity are kept and re-read on v_{top} . The Woodbury–Schur rewrite $-\text{sch} = \beta - \alpha$ is a finite SATZ, over $\mathbb{Q}$ and at machine precision, once $\alpha = (\eta - 1)/\kappa$ and $\beta = r^T K_W^{-1} r / \kappa$ with the r405 factor $\kappa = \Delta / (-\text{sch})$. It does not depend on which bulk vector is used as ℓ : the living/dead census $\beta > \alpha$ / $\beta < \alpha$ is exactly $\text{sch} < 0$ / $\text{sch} > 0$ (r401), for v_{top} and for the constant. The formula v_{top} is *not* the bulk defect ($\cos(v_{\text{top}}, v_{\text{SV}}) = 0.68$ at $k_z = 9$, 0.058 at $k_z = 18$ where $\|\mathfrak{T}_0\| = 1.25$). Euler-on- v_{top} is the constant content $|\langle 1, v_{\text{top}} \rangle|^2$, not α_T . No RH claim.

Status. Lemmas 1–3 are proved (finite identities). Propositions 1–3 are finite machine checks. Ausgang CHART_IDENTITY_EXACT / TOP_MODE_NOT_DEFECT. No L^* claim. No $R^\dagger$ claim. No RH claim.

1 Definitions

On the hole nodes Y let $u^\vee$ be the dual weight of r356/r407 and $P_Y(x) = \prod_{y \in Y} (x - y)$. The barycentric hole-top mode in the $\sqrt{u^\vee}$ -weighted sample coordinate is

$$v_{\text{top},i} \propto \frac{1}{\sqrt{u^\vee(y_i)} P'_Y(y_i)}. \quad (1)$$

Write $A_0 = R_{N-3} - \frac{1}{2}I$ on Y and U for the two last dual CD columns (r367). The ones-split of r405 with $\ell = v_{\text{top}}$ (normalised) is $H = A_0 + \ell \ell^T$,

$$\eta = \ell^T H^{-1} \ell, \quad r = U^T H^{-1} \ell, \quad K_W = I + U^T H^{-1} U, \quad \Delta = 1 - \eta + r^T K_W^{-1} r.$$

Set $\alpha_{\text{raw}} = \eta - 1$, $\beta_{\text{raw}} = r^T K_W^{-1} r$, so $\Delta = \beta_{\text{raw}} - \alpha_{\text{raw}}$. The r401 scalar sch is the last Schur complement of the mixed 3×3 Φ (two CD directions plus the Sherman–Morrison border). With $\kappa_{\text{emp}} = \Delta / (-\text{sch})$ the chart-scaled coordinates are $\alpha = \alpha_{\text{raw}} / \kappa_{\text{emp}}$ and $\beta = \beta_{\text{raw}} / \kappa_{\text{emp}}$. Independently, the T0-Rayleigh $\alpha_T = v_{\text{top}}^T (\mathfrak{T}_0^* \mathfrak{T}_0 - I) v_{\text{top}}$ is the candidate “defect energy after bulk continuation”.

2 Finite identities

Lemma 1 (Euler tail). *For every z and every integer $K \geq 0$, $1 - z^{K+1} = (1 - z) \sum_{r=0}^K z^r$. Over $\mathbb{Q}$, at $z = 1/3$ and $K = 5$, both sides equal $728/729$. In particular $E_p + T_p = 1$ pointwise on Y for each geometric prime factor.*

Lemma 2 (disk Parseval). *For $0 < q < 1$ and $K \geq 0$, $(1 - q) \sum_{r=0}^K q^r = 1 - q^{K+1}$. At $q = 1/2$ and $K = 6$ the reserve q^{K+1} is $1/128$.*

Lemma 3 (chart identity). *On the r405 rational 3×3 ($H = \text{diag}(2, 3, 4)$, $\ell = (1, 1, 1)$, $U = I$, $J^{-1} = \text{diag}(1, 1, 2)$) one has $\eta = 13/12$, $\Delta = 7/36$, $\text{sch} = 21/8$, $\kappa = -2/27$, and with the definitions of §1*

$$\alpha = -\frac{9}{8}, \quad \beta = -\frac{15}{4}, \quad \beta - \alpha = -\text{sch}$$

exactly over $\mathbb{Q}$. On every measured source window the same rewrite holds at machine precision ($|\beta - \alpha - (-\text{sch})| \leq 2 \cdot 10^{-17}$) for $\ell = v_{\text{top}}$ and for $\ell = 1$.

Remark 1. Lemma 3 is determinant / congruence bookkeeping in the r401 Sylvester chart plus the r406 Woodbury form. It does *not* produce a new sign criterion: $\beta > \alpha$ if and only if $\text{sch} < 0$, independently of the bulk vector ℓ used to form H . The closed formula $\kappa = -(1 - \eta) \det K_2 / \det K_W$ of the 3×3 toy does not match κ_{emp} on the source two-column split (that U is not the U of Φ). $\kappa = 1$ is false: $\Delta = O(10^{-3})$ while $|\text{sch}| = O(10^{-2})$.

3 What the Euler tail sees on v_{top}

Proposition 1 (Euler re-read). *Because $E_p + T_p = 1$ pointwise, $\sum_p \nu_p |\langle E_p + T_p, v \rangle|^2 = |\langle 1, v \rangle|^2$ for every v . At $k_z = 9$, this equals 0.01192 on v_{top} and is not $\alpha_T = -0.07302$. The geometric Euler span misses v_{top} (QR-relres 0.74). A positive source representation of $\beta - \alpha$ by Euler terms plus disk Parseval on v_{top} is therefore not available. The constant-vector control reproduces the r405 pins ($\Delta = 2.110 \cdot 10^{-3}$, $c_2 = 0.99789$, $\kappa = 0.03151$).*

Proposition 2 (not the defect). *v_{top} is not $\mathfrak{T}_0$'s top singular vector and not A_0 's negative eigenvector. At $k_z = 9$, $\cos = 0.6766$ and $\alpha_T < 0$ (contractive); the true top singular value has Rayleigh $+0.16670$. At $k_z = 18$ (a genuine one-defect window, $\|\mathfrak{T}_0\| = 1.25$) the cosine drops to 0.058. $H = A_0 + v_{\text{top}} v_{\text{top}}^T$ stays indefinite on 24/42 MAIN windows (PD on only 18/42). The barycentric Y -top mode and the min-norm interpolant's excess direction are different orthogonalities (discrete OP on the holes versus dual OP chain of the full measure, evaluated on Y).*

4 Census and kills

Proposition 3 (sign census). *On the named census, with the chart-scaled (α, β) of §1: MAIN 42/42 have $\beta > \alpha$ (all $\text{sch} < 0$); living χ_3 37/37 and living χ_4 41/41 (together 78) have $\beta > \alpha$; dead χ_3 5/5 and dead χ_4 -20 (together 6) have $\beta < \alpha$. Death remains $\text{sch} > 0$ (r401), not a top-mode under-coverage beyond that tautology. Cosines $\cos(v_{\text{top}}, v_-)$ range in $[0.006, 0.816]$.*

Kills, all of which fire. (1) The constant reproduces r405 (wrong vector, as designed). (2) Dropping $\sqrt{u^\vee}$ ($v \sim 1/P_Y'$) drops the $k_z = 9$ singular-vector cosine $0.68 \rightarrow 0.38$. (3) Replacing P_Y' by $y_i - \text{mean}(Y)$ drops it to 0.17. (4) Weight-permute has $n_-(I - \mathfrak{T}^* \mathfrak{T}) = 20$; scramble has 21. The bulk is already dead; an edge balance of one mode is then irrelevant. (5) Omitting the Sherman–Morrison border leaves the mixed 2×2 K_2 with $\text{ind}_- = 1$ on *both* living $k_z = 9$ and dead χ_3 -15; without the border there is no living/dead scalar.

5 What this is not

This round does not prove HTM, QD, or P1. It does not claim $R^\dagger \succ \frac{1}{2}I$. It does not move the mincut (`selected_augDualResolvent_gt_half`, base 4 / refined 5). The edge theorem asked for in DCCLXXIX, as a *top-mode-specific* positivity $\beta > \alpha$ with a positive Euler source, is reduced to the r401 chart plus the r405/r406 Woodbury rewrite. Whether HTM itself holds is the parallel round 413. No RH claim.